NG PULSE

Rail-Signature Workload Integrity Monitor

EEL 4914 Senior Design I
Divide and Conquer Proposal Draft

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Executive Summary

NG PULSE is a hardware-security monitoring project intended to detect physical attacks against a protected processor by observing its supply-rail current behavior. The proposed monitor will attach to any printed circuit board (PCB), acquire current signatures from the host processor, compare measured behavior against a known-good reference or golden profile, and generate an alert when an abnormal condition is detected. An advanced implementation may also initiate a protective system shutdown after an attack is identified.

The planned architecture combines a current-sense front end, a 12-bit analog-to-digital converter (ADC), windowed feature extraction, and threshold or classification logic. The project will first establish reference current waveforms for selected RISC-V processor functions or workloads. The detection logic will then be implemented in Verilog and verified before physical prototyping. The prototype is planned as a PCB containing the current-sense design and MCU for processing, to monitor a ESP32-C (board with a RISC-V processor) during normal function.

The project is funded by Northrop Grumman with a maximum budget of $1,000. Key performance targets include 100% coverage of the reference waveforms defined by the project test plan, no more than a 5% reduction in RISC-V computing efficiency, no more than 1 second of attack-detection latency, at least a 95% attack-detection rate, no more than a 5% false-positive rate, 12-bit ADC resolution, and operation from either a 3.3 V or 5 V input supply.

1. Motivation and Background

Modern computing systems depend on integrated circuits that may be exposed not only to software-based threats but also to physical attacks that interfere directly with hardware operation. NG PULSE is motivated by the need for an additional layer of hardware-level security that can identify abnormal electrical behavior in a protected processor. A representative threat identified for this project is abnormal or high-current injection intended to disturb normal chip operation.

The central concept is to monitor the processor's supply-rail current signature. Under normal operation, processor activity produces characteristic electrical behavior. By establishing a golden representation of that behavior and continuously comparing later measurements against it, NG PULSE is intended to identify departures that may indicate an attack. Because the monitor does not need to sit in the processor's normal control path, the design goal is to provide security monitoring while limiting the impact on host computing performance.

The project also provides a mixed-signal design flow spanning analog sensing, analog-to-digital conversion, digital signal processing, classification or threshold logic, Verilog/RTL development, CAD-based schematic integration, and eventual

physical prototyping. The final design is intended to be suitable for attachment to an existing PCB rather than requiring a complete redesign of the protected system.

2. Goals and Objectives

2.1 Goals

Basic Goals:

- Develop a functional hardware-security monitor capable of detecting abnormal supply-rail current behavior from an ESP32-C target and asserting an error flag when an abnormal condition is detected.
- Monitor one predefined ESP32-C work mode by comparing measured supply-rail current behavior against an established golden profile representing normal operation.
- Monitor the ESP32-C while in a Not-in-Use state to detect current behavior indicative of unauthorized processor activity.
- Implement NG PULSE as a separate monitoring system that can interface with an external target without requiring the monitoring hardware to be integrated into the target PCB.
- Demonstrate detection of abnormal supply-rail current behavior resulting from a defined physical/electrical attack on the ESP32-C.
- Produce digital design of ASIC in Cadence to contain comparison logic and error flag generation

Advanced Goals:

- Expand NG PULSE from monitoring one predefined work mode to monitoring four predefined ESP32-C work modes in addition to the Not-in-Use state.
- Automatically apply the appropriate golden profile according to the operating mode reported by the ESP32-C.
- Maintain continuous monitoring as the ESP32-C transitions between its predefined operating modes and the Not-in-Use state.

Stretch Goals:

- Integrate the current-sensing hardware, Nordic MCU, and FPGA components onto a single custom NG PULSE PCB while maintaining the ESP32-C as an externally connected target.
- Incorporate Bluetooth connectivity to provide system status and detection information to a mobile device.

- Demonstrate a protective response in which the NG PULSE error flag can initiate an automatic shutdown of the ESP32-C following detection of abnormal current behavior.
- Include ADC functionality in digitally designed ASIC
- Include memory storage in digitally designed ASIC
- Include current capture in digitally designed ASIC

2.2 Objectives

Basic Objectives:

- Design and implement a shunt-based current-sensing front end on a custom NG PULSE PCB to convert ESP32-C supply-rail current behavior into an analog voltage signal measurable by the Nordic MCU's integrated ADC.
- Connect the Nordic MCU, FPGA, and ESP32-C target to the NG PULSE current-sensing PCB through external jumper connections.
- Characterize one predefined authorized ESP32-C work mode by collecting 1,024 current measurements using the Nordic MCU's integrated ADC.
- Use the 1,024 characterization measurements to calculate an average current value and acceptable deviation representing the golden profile for the predefined work mode, and store the resulting profile on the Nordic MCU.
- Establish communication from the ESP32-C to the Nordic MCU so that the target can report whether it is operating in the predefined work mode or the Not-in-Use state.
- Configure the Nordic MCU to acquire current measurements and provide the FPGA with the measured ADC values and stored golden-profile parameters required for detection.
- Develop Verilog-based FPGA logic that evaluates successive current measurements against the acceptable range established by the golden profile and asserts an error flag when abnormal current behavior satisfies the defined detection criteria.
- Use developed RTL code to produce a logic gate schematic in Cadence
- Continue monitoring supply-rail current while the ESP32-C reports the Not-in-Use state and assert an error flag when current behavior indicates unauthorized processor activity.
- Simulate and verify the Verilog-based comparison, detection, and error-flag logic prior to physical FPGA implementation.
- Physically demonstrate that NG PULSE can detect abnormal ESP32-C supply-rail current behavior resulting from an induced voltage-surge attack and successfully assert an error flag.

Advanced Objectives:

- Characterize four predefined ESP32-C work modes using 1,024 current measurements per mode and establish a separate golden average and acceptable deviation for each mode.
- Store the golden-profile parameters for all four predefined work modes on the Nordic MCU.
- Expand ESP32-C-to-Nordic communication to identify the Not-in-Use state and each of the four predefined work modes.
- Configure the Nordic MCU to automatically select the appropriate stored golden profile according to the operating mode reported by the ESP32-C.
- Provide the FPGA with the applicable golden-profile parameters and current ADC measurements as the ESP32-C transitions between predefined operating modes.
- Expand FPGA-based detection to evaluate abnormal current behavior across all four predefined work modes while maintaining monitoring of the Not-in-Use state.

Stretch Objectives:

- Develop a custom NG PULSE PCB incorporating the current-sensing circuitry, Nordic MCU, and FPGA components on a single board, eliminating the need for a separate FPGA board while maintaining an external connection to the ESP32-C target.
- Utilize the Nordic MCU's Bluetooth capability to communicate the ESP32-C's reported operating mode and NG PULSE's normal, abnormal, or error status to a mobile device.
- Implement and demonstrate an automatic ESP32-C shutdown response initiated by the FPGA-generated error flag as an example of a protective action that can be taken following detection.
- Design a 4-bit ADC digitally
- Expand ADC resolution to 12 bits and integrate into ASIC
- Design memory storage, access, and data transfer system for ASIC
- Design voltage amplification and straight through power rail for current sense in ASIC

3. System Features and Functional Description

NG PULSE will observe the supply-rail current of the protected processor and convert the sensed analog behavior into digital data. The digital subsystem will process the measurements over defined time windows and compare extracted information against the golden reference profile. When the measured behavior satisfies the project's criteria for an anomalous or attack condition, an attack flag will be asserted.

- Supply-rail current sensing.

- 12-bit analog-to-digital conversion.
- Acquisition and storage/use of golden reference waveforms.
- Windowed feature extraction or equivalent waveform-processing method.
- Threshold-based or classifier-based anomaly decision logic.
- Attack flag generation within the specified latency.
- Alert/status reporting; PMBus or I²C is identified in the project concept, with the final interface to be confirmed.
- Optional system shutdown response for the stretch implementation.
- Operation from either a 3.3 V input supply.

4. Existing Products and Prior Related Work

The NG PULSE project overlaps with several established areas of technology: current-sense amplifiers, ADC-based power monitors, threshold and comparator protection circuits, processor power-signature analysis, and physical fault-injection security research. Although each of these areas provides technologies relevant to the proposed system, the information currently provided for this report does not establish a specific commercial product that directly combines these capabilities. The following products were therefore selected as comparable due to their demonstration of individual functions that overlap with the proposed NG PULSE architecture. The differences that will separate the NG PULSE from these systems are also discussed.

4.1 Texas Instruments INA233

The Texas Instruments INA233 is a ten-pin Very Thin Shrink Small Outline Package (VSSOP) digital current-sense monitor designed to measure and report electrical signals from a monitored power rail. The INA233 takes measurements of the current, voltage, and power and can report these values to enable accurate decision making in controlled systems. It has an integrated analog-to-digital converter (ADC) with programmable conversion times and averages modes. This device also has programmable limits for voltage, current, and power that can send alerts when defined thresholds are exceeded. This combination of features allows for continuous monitoring with a notification system for abnormal behaviors [1].

The INA233 shares several characteristics with the NG PULSE. The integration of current sensing, the analog-to-digital conversion, digital processing, threshold detection, and notification system. Both of these components are designed to be placed directly onto a programmable circuit board to enable the pieces to monitor a power rail so that abnormal conditions can be mitigated. The difference comes from the recognition of abnormal behaviors. The INA233 has programmable

limits that will send an alarm if reached. The NG PULSE will analyze the current during processor activity to identify abnormal behavior.

4.2 Analog Devices LTC2945

The Analog Devices LTC2945 is a twelve-pin Mini Small Outline Package (MSOP) power monitor that is designed to monitor the electrical characteristics of a power rail. This device can measure current, voltage, and power to store these values and send the measurements through a digital interface. The measurements taken can be adjusted to either be singular or continuous. Similar to the INA233, this device has programmable limits for detecting what is deemed to be abnormal behaviors. The LTC2945 also includes a shutdown mode that can reduce the current to 20 microamps, which reduces power consumption when available [2].

The LTC2945 is comparable to NG PULSE due to both systems providing power-rail monitoring, current and voltage measurement, threshold based detection, and a communication system for results. The difference between these two comes from the primary purpose. The LTC2945 is primarily for power-management by measuring power consumption, whereas the NG PULSE is behavioral due to examining current behavior against a device's golden-profile.

4.3 Analog Devices ADM1293

The Analog Devices ADM1293 is a fourteen-pin Thin Shrink Small Outline Package (TSSOP) power monitor that monitors a power rail to take measurements. The device has an integrated 12-bit ADC to measure voltage to determine the load current, and is capable of calculating power from the measured values. The measurement modes can be adjusted to either singular or continuous, and can take averages for stability and reduction of sensing noise. The programmable thresholds within this device can identify abnormal behavior that triggers an alert to notify the system [3].

The ADM1293 has similarities to the NG PULSE in that both systems combine current sensing, an ADC, configurable sampling, threshold detection, and alert generation. Both devices can also store measurements and identify abnormal behavioral conditions. The difference comes from the ADM1293 is designed for general power faults instead of the NG PULSE's focus of the processor's current signature. Instead of limiting detection to overcurrent or overvoltage, NG PULSE analyzes the characteristics of the current waveform and uses the measurements to determine whether the processor is operating within its normal behavior.

5. Engineering Specifications

The following specifications define the current quantitative targets for the NG PULSE system. The final test plan should define the exact workloads, attack cases, measurement equipment, and pass/fail procedures used to demonstrate each requirement.

Table 5.1: Engineering Specifications

Parameter	Requirement	Verification Method
Reference waveform coverage	# of target processes: 4 Sample requirement of each process: 1024 samples averaged Output: golden profile	Golden-profile acquisition records
RISC-V Computing Efficiency	0% loss (zero tolerance with NG PULSE operation)	Benchmark with/without monitor
Attack detection latency	Maximum 1 s from detectable attack/anomaly to asserted flag	Timed attack/flag measurement
Attack detection rate	Minimum 95%	Repeated defined attack trials
False-positive detection rate	Maximum 5%	Repeated normal-workload trials
F1 score	Minimum 0.85	Repeated and defined attack trials and normal operation monitoring
Current sensing range	500mA to 20A	Thermal imaging camera under high load and minimum current detection within specified accuracy.
Signal to noise ratio	Minimum 20dB	Golden profile comparison to actual waveform
Long time overcurrent shutoff	< 20s for 1.05 - 1.3 pu surge	Repeated and defined attack trials
Short time overcurrent shutoff	< 5s for 1.3 - 2.0 pu surge	Repeated and defined attack trials
Instantaneous overcurrent shutoff	< 0.1s for >= 2.0 pu surge	Repeated and defined attack trials

Weight	< 5 lbs	Digital scale measurement of complete system
Size	< 1 ft^2	Physical dimension and footprint measurement
Cost	< $1000 (Shipping not included)	BOM and purchase receipt cost calculation

6. House of Quality

A formal House of Quality will be developed to connect sponsor/customer needs to measurable engineering characteristics. Based on the current project definition, likely customer needs include reliable attack detection, low false alarms, rapid response, minimal performance impact, compatibility with existing hardware, and practical prototyping cost.

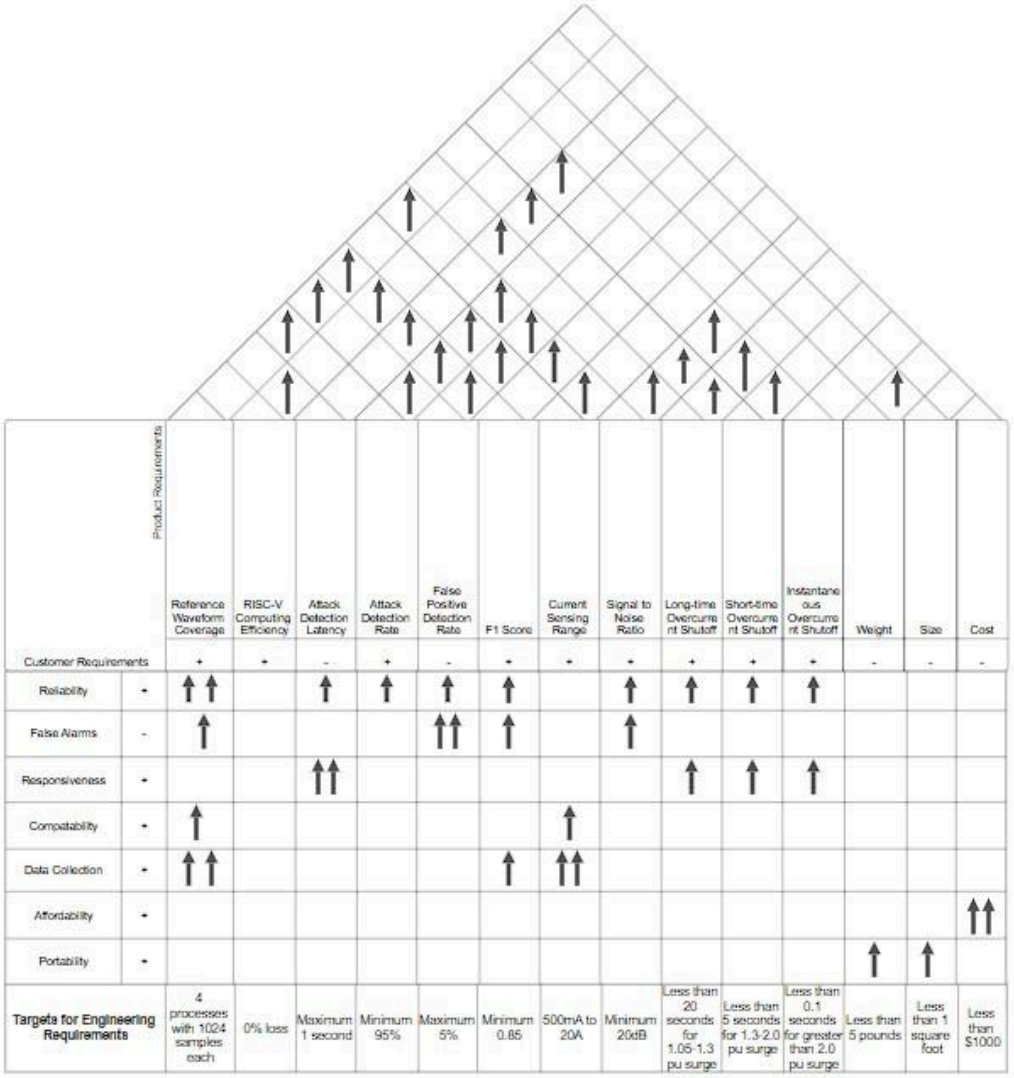


Figure 6.1 - House of Quality

7. Proposed Hardware Architecture

At a high level, the protected processor is powered through a rail that is observed by the NG PULSE sensing circuitry. The current-sense front end produces a measurable analog representation of the rail current. A 12-bit ADC digitizes this signal. Digital processing then extracts waveform information and compares the measured behavior against the golden profile. The decision engine asserts an attack flag when the anomaly criteria are satisfied. The flag may be reported externally and, in the advanced implementation, may initiate a shutdown response.

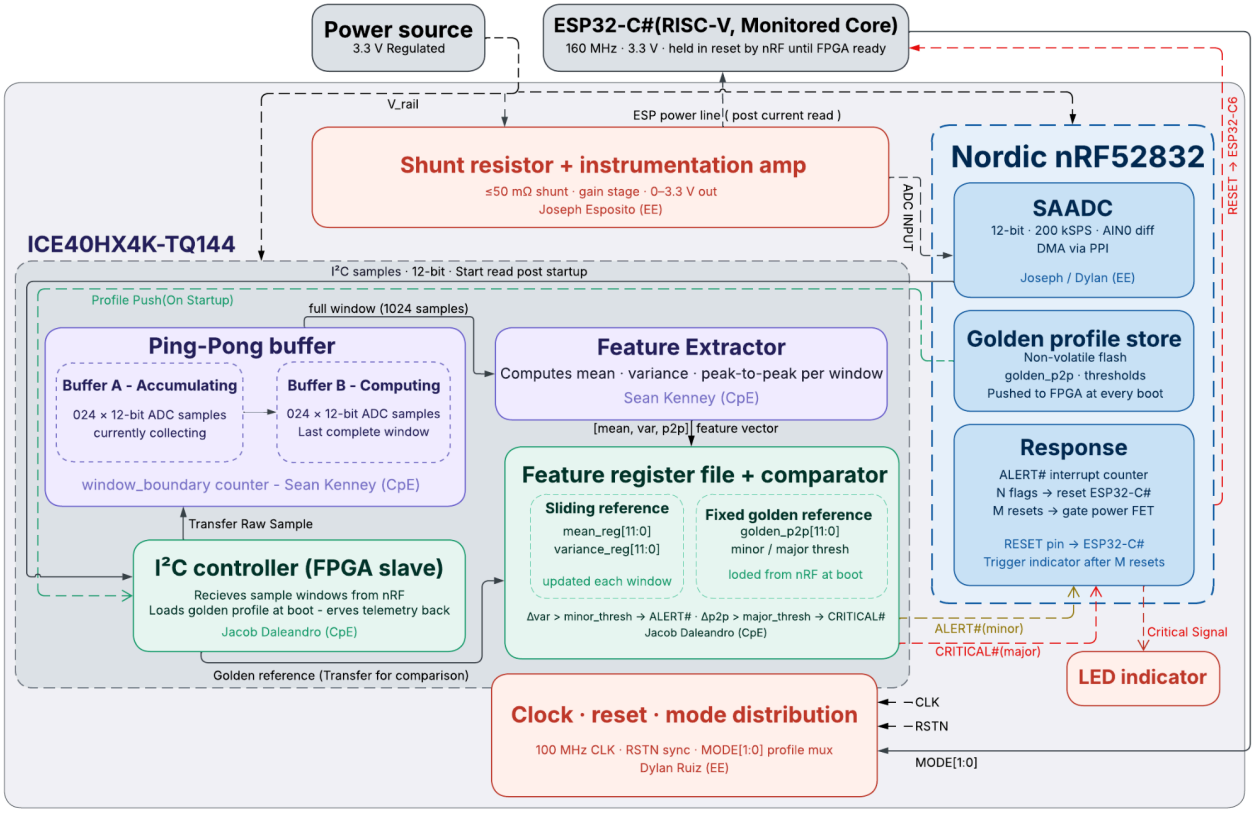


Figure 7.1: Hardware Block Diagram

7.1 Preliminary Component Categories

- Shunt resistor
- Op-amp
- MCU: Nordic nrf52832
- FPGA: ICE40HX4k-TQ144
- Target RISC-V processor/platform: ESP32-C
- Communication/interface components:TBD
- Protection/shutdown circuitry: MOSFET

8. Digital Logic and Software/RTL Flow

The NG PULSE digital subsystem will be developed in Verilog. The intended processing flow begins with digitized current samples from the ADC. Samples will be organized into appropriate analysis windows, processed to obtain the features or values required by the detector, and evaluated against the golden profile or configured decision thresholds. When the decision logic identifies an attack condition, it will assert the attack flag and provide the required reporting or protection response.

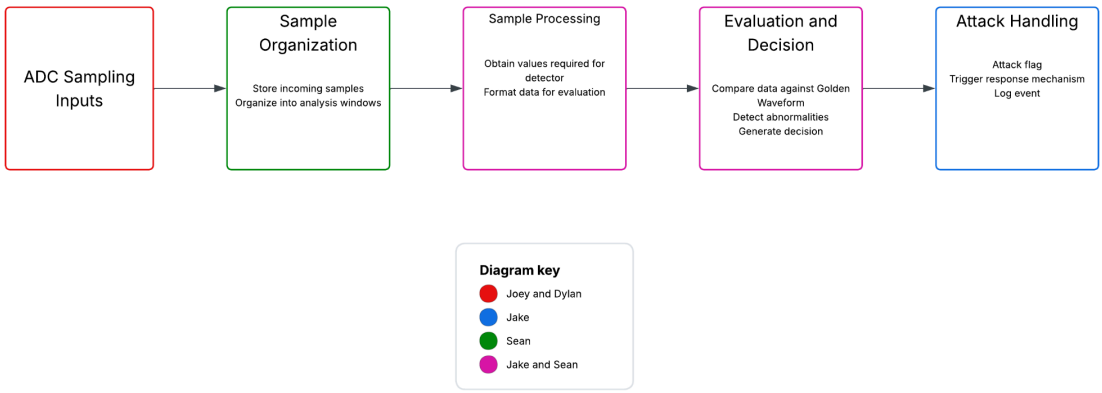


Figure 8.1: Software Flowchart

9. Prototype Plan

The initial physical detector will be implemented on a PCB using commercial off-the-shelf components. The PCB will contain the sensing, controls, and clock signals, as well as pin headers for the ESP32-C, FPGA, and Nordic board, and supporting power circuitry required to evaluate the NG PULSE concept.

The preferred prototype will connect the detector to an FPGA available in the Senior Design laboratory so that the Verilog logic can be implemented, tested, and modified rapidly. The team will connect the prototype to a RISC-V processor board (ESP32-C) in the DRACO laboratory. This would allow the monitor to acquire and evaluate current signatures from an actual RISC-V target.

Prototype testing will verify current acquisition, golden-profile comparison, attack flag generation, detection latency, detection rate, false-positive rate, and all other specifications listed in table 5.1.

10. Budget and Financing

NG PULSE is funded by Northrop Grumman with a maximum project budget of $1,000. The team will use existing UCF Senior Design and DRACO laboratory equipment where available to reduce costs. The project bill of materials will be updated as exact components are selected.

Table 10.1: Cost Breakdown

Budget Item	Example Contents	Estimated Cost	Status
Current sensing	Sensor/amplifier, shunt, passives	[TBD]	To be selected
Data conversion	12-bit ADC and support components	[TBD]	To be selected
Detection/support	Comparator, logic/support ICs	[TBD]	To be selected
Power	Regulators, protection, filtering	[TBD]	Investigating
Prototype materials	Breadboard, wiring, connectors, headers	[TBD]	Planned
PCB/fabrication	Prototype PCB and assembly as required	[TBD]	Future
Contingency	Replacement parts/shipping/unplanned needs	[TBD]	Reserved
TOTAL	Must remain within sponsor funding	≤ $1,000	Requirement

11. Project Milestones

11.1 Senior Design I

SD1 will focus on establishing the technical foundation needed to begin prototyping immediately in SD2. By the end of SD1, the team intends to have all major hardware selected, the golden profile obtained, a schematic of the proposed ASIC/system completed, and at least 80% of the Verilog code complete.

Table 11.1: SD1 Milestones

SD1 Milestone	Target	Completion Evidence
System architecture and requirements	10/20/2026	Approved architecture/specifications
Hardware component selection	11/6/2026	Selected Parts/BOM
Hardware schematic V1	11/13/2026	Fusion360 screenshots

Verilog code major functions determined (not written)	11/13/2026	Verilog code files
Hardware schematic V2	11/20/2026	Fusion360 screenshots
High-risk hardware testing complete	11/20/2026	Test results logged, working hardware obtained if possible
Verilog development ≥80% complete	11/27/2026	Code files
PCB(s) laid out in Fusion360, ready to order	11/27/2026	Fusion360 screenshots

11.2 Senior Design II

Physical prototyping will begin immediately at the start of SD2. Initial work will focus on constructing the breadboard detector, integrating the FPGA and RISC-V target, and establishing basic end-to-end operation. The remaining approximately two months of SD2 will be devoted primarily to troubleshooting, refinement, final specification testing, data collection, and preparation for the final demonstration and presentation.

Table 11.2: SD2 Milestones

SD2 Milestone	Target	Completion Evidence
Complete soldering for PCB(s)	1/16/2027	Complete PCB ready for testing
Complete Verilog code(s) ready to upload	1/16/2027	Code files
Test current capture accuracy	1/23/2027	Lab multimeter
Use RTL to design ASIC	1/30/2027	Cadence cloud files
Acquire golden profile and begin full system testing	1/30/2027	MCU Memory, system test results
Troubleshooting period	1/30 - 4/30/2027	Recorded pass/fail results for key specifications

12. Team Organization and Work Distribution

Table 12.1: Work Distribution

Team Member	Primary Responsibility	Secondary Responsibility
Joey Esposito	ADC and supporting circuitry selection, comparator/protection circuitry, power distribution requirements and hardware component/BOM selection.	Hardware schematic development, FPGA/processor hardware interface, prototype assembly, electrical verification and specification testing
Dylan Ruiz	Current-sense design, current-sensor/amplifier selection, shunt/passive component selection, analog signal conditioning and power/voltage regulator design	Hardware schematic development, analog/ADC interface verification, breadboard prototype construction, hardware testing and construction
Sean Kenney	RISC-V/workload simulation, golden model and golden profile generation, windowed feature extraction RTL and Verilog module development	RTL testbenches, simulation/data analysis, ADC to digital logic integration, FPGA implementation and debugging
Jacob Daleandro	Threshold/classifier engine, attack detection and attack flag logic, PMBus/I²C/status interface logic and Verilog module development	Golden-profile comparison logic, RTL verification, FPGA/RISC-V integration, digital-system testing and debugging

Both Dylan and Joey are electrical engineering majors and as such will select all hardware for this project together, including, but not limited to the current capture circuit, ADC, and voltage regulator. They also will be responsible for producing the schematic and verifying all power and connection requirements. Sean and Jacob are both computer engineering majors so they will be creating a virtual model of the RISC-V processor and obtain the golden profile from their

simulation. They will write the verilog code to program the FPGA and Nordic board as well.

13. Technical Risks and Open Decisions

- The exact definition of the physical attack(s), including the high-current injection test method and safe laboratory limits, must be established with the sponsor/advisor.
- The exact golden-profile generation method, waveform feature set, and threshold/classifier algorithm remain to be finalized.
- The shutdown mechanism and interface to the protected system remain a stretch design decision.
- Exact component part numbers, sampling rate, sensor range/bandwidth, and detailed analog performance requirements remain to be selected.
- This project and its specifications are determined in large part by the team at Northrop Grumman and so they are subject to change depending on their direction.

14. Preliminary Verification and Test Strategy

Verification will proceed from golden-profile acquisition through virtual RTL verification and finally to breadboard prototype testing. Normal workloads will be used to establish the reference set and evaluate false-positive behavior, while defined attack trials will be used to evaluate detection rate, latency, F1 score, and shutdown functionality. Processor benchmarks will be run with and without NG PULSE attached to determine the impact on computing efficiency. Current-sensing performance and signal quality will also be evaluated to ensure that the acquired current waveforms are suitable for reliable anomaly detection.

- **Reference-waveform test:** For all four (4) selected processes, collect and average 1024 current samples to establish each golden profile. Verify that a golden waveform is successfully acquired for each selected process.
- **Efficiency test:** Run a defined RISC-V benchmark with and without NG PULSE attached. Pass if there is no reduction in computing efficiency (0% efficiency loss).
- **Latency test:** Measure the time from a detectable attack/anomaly to assertion of the alert flag. Pass if the detection latency is ≤1 s.
- **Detection-rate test:** Run repeated, defined attack trials. Pass if ≥95% of attacks are correctly detected and flagged.
- **False-positive test:** Run repeated normal-workload trials. Pass if ≤5% of normal trials are incorrectly identified as attacks.

- **Current-sensing range test:** Verify that the current-sensing system operates over the specified range of 500 mA to 20 A. Minimum-current detection will be verified within the specified accuracy, and a thermal imaging camera will be used under high-load conditions to verify safe operation.
- **Signal-to-noise ratio (SNR) test:** Compare acquired current waveforms to the established golden profile to evaluate signal quality relative to measurement noise. Pass if the measured SNR is ≥20 dB.
- **F1 score test:** Run repeated defined attack trials and normal-operation trials and calculate the F1 score from the resulting detection performance. Pass if the F1 score is ≥0.85.
- **Long-time overcurrent shutdown test:** Apply a sustained current surge of 1.05–1.3 pu and verify that system shutdown occurs within 20 s.
- **Short-time overcurrent shutdown test:** Apply a current surge of 1.3–2 pu and verify that system shutdown occurs within 5 s.

15. Conclusion

NG PULSE proposes a hardware-security monitor that uses processor supply-rail current behavior as an indicator of workload integrity and abnormal physical activity. The project will establish golden current profiles for four selected processor processes, implement current sensing and digital detection hardware, verify the design virtually, and build a physical prototype for evaluation. The system will be evaluated using measurable performance criteria including detection latency, attack detection rate, false-positive rate, F1 score, signal-to-noise ratio, current-sensing range, processor efficiency impact, and overcurrent shutdown response.

The project is structured so that SD1 produces a prototype-ready design and SD2 begins immediately with integration and testing, followed by an extended troubleshooting and verification period. The current project definition provides measurable system targets and a clear development path, while several details remain intentionally open for team, sponsor, and advisor decisions. These details should be finalized as component selection, laboratory access, attack definitions, team assignments, and detailed design work progress.

Appendix A - References

Course guideline: EEL 4914 Senior Design I, Divide and Conquer Guideline. Additional technical references and product datasheets must be added as the prior-work review and component selection are completed.

[1] Texas Instruments “SBOS790 Data sheet | TI.com.” https://www.ti.com/document-viewer/INA233/datasheet

[2] Analog Devices, Inc., *“LTC2945: Wide Range I²C Power Monitor,”* Rev. D. https://www.analog.com/media/en/technical-documentation/data-sheets/ltc2945.pdf

[3] Analog Devices, Inc., *“ADM1293/ADM1294: Digital Power Monitor with PMBus Interface,”* Rev. B, 2015. https://www.analog.com/media/en/technical-documentation/data-sheets/ADM1293_1294.pdf

Appendix B - Copyright Permissions

Appendix C - Supporting Material